SRINIVAS GAMPASANI

Generative AI Engineer | ML Engineer | Data Scientist
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SUMMARY

- Results-driven **Generative AI Engineer** with **4+ years of experience** in **Machine Learning, Deep Learning, and Artificial Intelligence**, specializing in **Large Language Models (LLMs)**, **Agentic AI**, and **RAG-based systems** across healthcare and enterprise environments.
- Skilled in fine-tuning and deploying **GPT, LLaMA, BERT, and Transformer-based architectures** using **HuggingFace Transformers, LangChain, LangGraph, and OpenAI APIs** to build advanced clinical decision support, text generation, and multimodal AI applications.
- Proficient in **Retrieval-Augmented Generation (RAG), GraphRAG, multi-agent orchestration**, and **LLM evaluation** using RAGAS, TruLens, and Guardrails AI.
- Experienced in deploying scalable AI pipelines using **Docker, Kubernetes, and CI/CD** across **AWS, Azure, and GCP**, and managing **LLMOps and MLOps workflows** with **MLflow, Kubeflow, and ArgoCD**.
- Adept at integrating **Vector Databases (FAISS, Pinecone, Weaviate)** for semantic search and RAG, **Knowledge Graphs (Neo4j)** for context-aware reasoning, and optimizing models with **LoRA, QLoRA, PEFT, vLLM, and ONNX** for cost-efficient inference.
- Proven impact: **40% improvement in clinical answer relevance, 60% compute cost reduction, 35% hallucination reduction**, and 30% faster deployments. Committed to building **HIPAA-compliant, responsible, and explainable AI** solutions.

KEY SKILLS

Generative AI & LLMs: GPT-4/Turbo, LLaMA, Gemini, Claude, Mixtral, BERT, T5, RAG, GraphRAG, LLM Fine-Tuning (LoRA, QLoRA, PEFT), Instruction Tuning, RLHF, DPO, Prompt Engineering, Chain-of-Thought, Tool Calling, OpenAI API
Agentic AI: LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, MCP (Model Context Protocol), Multi-Agent Orchestration, Supervisor-Worker Patterns, Human-in-the-Loop, Autonomous Task Decomposition
LLM Evaluation & Responsible AI: RAGAS, TruLens, DeepEval, Guardrails AI, Nemo Guardrails, SHAP, LIME, Hallucination Mitigation, Fairness & Bias Mitigation, Explainable AI, HIPAA Compliance, AI Governance, Datadog, Terraform
NLP & Computer Vision: HuggingFace Transformers, Sentence Transformers, spaCy, NER, Summarization, Semantic Search, Hybrid Search, BM25, Reranking, CNNs, Vision Transformers (ViT), OpenCV, CLIP, YOLO, Diffusion Models, SAM
ML & Deep Learning: PyTorch, TensorFlow, JAX, Scikit-learn, XGBoost, LightGBM, Neural Networks, LSTMs, Random Forest, Ensemble Methods, Anomaly Detection, Time Series, Feature Engineering, Hyperparameter Tuning, Quantization, Distillation, ONNX, vLLM, Weights & Biases, Multimodal Models, Azure OpenAI, AWS Bedrock
MLOps & LLMOps: MLflow, ClearML, Kubeflow, DVC, Docker, Kubernetes, Helm, ArgoCD, GitOps, CI/CD, Jenkins, GitHub Actions, FastAPI, Flask, gRPC, GraphQL, Model Monitoring, Drift Detection, Prometheus, Grafana, OpenTelemetry
Data Engineering & Big Data: Apache Spark, Apache Kafka, Apache Airflow, Apache NiFi, dbt, Delta Lake, Apache Iceberg, ETL/ELT Pipelines, Feature Stores, Data Lakes, Distributed Computing, PySpark, Trino, ClickHouse
Data Warehousing & Databases: Snowflake, Databricks, BigQuery, AWS Redshift, PostgreSQL, MongoDB, Neo4j, Redis, Elasticsearch, FAISS, Pinecone, Weaviate, Milvus, Chroma, Azure AI Search
Cloud & AI Services: AWS (SageMaker, Bedrock, Glue, Athena, S3, Lambda, EKS, EC2, EMR), Azure (ML Studio, OpenAI, AI Search, Databricks, Data Factory, Copilot Studio), GCP (Vertex AI, BigQuery, ADK, Agent Space, Cloud Composer)
Programming Languages: Python, SQL, PySpark, Java, Scala, TypeScript, JavaScript, R, Go, Bash
Visualization & BI: Power BI, Tableau, Streamlit, Plotly, Matplotlib, Dashboard Development, KPI Monitoring

PROFESSIONAL EXPERIENCE

Generative AI Engineer & Data Scientist | Ascension Via Christi Health | Wichita, KS | 09/2024 – Present
Ascension Via Christi Health is a Catholic-sponsored health system fully owned by Ascension Health, providing clinical care across hospitals, clinics, and specialty facilities in the United States.

- Architected **RAG and GraphRAG pipelines** using GPT-4, LLaMA, and Gemini via LangChain, LlamaIndex, Azure OpenAI, and Azure AI Search with hybrid dense + sparse + reranking retrieval, raising clinical answer relevance by **40%** and enabling real-time AI decision support for **500+ clinicians**.

- Engineered **multi-agent AI systems** using LangGraph with chain-of-thought prompting, autonomous tool calling, MCP integration, supervisor-worker patterns, and long-term memory across 5 specialized agents.
- Integrated **Guardrails AI and Nemo Guardrails** for LLM output moderation, cutting hallucinations by **35%** and enforcing Responsible AI policies across clinical workflows.
- Applied **LLM fine-tuning (LoRA, QLoRA, PEFT)**, instruction tuning, RLHF, DPO preference optimization, quantization, and distillation for clinical domain adaptation; evaluated with RAGAS and TruLens, reducing compute cost by **60%** via vLLM optimized inference.
- Designed **NLP pipelines** for clinical summarization, NER, text classification, and structured extraction from **50,000+ patient records** using BERT, Transformers, HuggingFace, spaCy, and Apache Spark; applied OCR and Knowledge Graphs (Neo4j) to reduce manual documentation effort by **30%**.
- Engineered scalable **ETL/ELT data pipelines** processing 50,000+ clinical records using Apache Spark, PySpark, SQL, Snowflake, and Delta Lake, reducing data processing time by **30%** through automated validation and monitoring.
- Implemented end-to-end **MLOps and LLMOps pipelines** with MLflow, Docker, Kubernetes (EKS), Helm, ArgoCD, CI/CD (Jenkins, GitHub Actions), and OpenTelemetry on AWS SageMaker and Azure ML Studio, achieving zero-downtime deployments and **30% fewer release errors**.
- Applied **SHAP and LIME** for model explainability, enforced HIPAA-compliant OAuth2 and RBAC data governance, and designed Power BI and Streamlit dashboards for KPI monitoring and stakeholder reporting.

Technologies Used: GPT-4, LLaMA, LangChain, LangGraph, LlamaIndex, Azure OpenAI, AWS SageMaker, FAISS, Pinecone, Neo4j, BERT, HuggingFace, Apache Spark, Snowflake, Delta Lake, MLflow, Docker, Kubernetes, Helm, ArgoCD, Jenkins, GitHub Actions, RAGAS, TruLens, Guardrails AI, SHAP, LIME, Power BI, Streamlit

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Machine Learning Engineer & Data Engineer | Colgate-Palmolive | Topeka, KS | 12/2023 – 08/2024

Colgate-Palmolive is an American multinational consumer products company with operations in over 200 countries, focused on oral care, personal care, and home care products.

- Built end-to-end **Machine Learning pipelines** for demand forecasting and predictive analytics using XGBoost, LightGBM, Random Forest, and Neural Networks (PyTorch, TensorFlow), achieving **20% prediction accuracy improvement** through feature engineering, hyperparameter tuning, A/B testing, and ensemble modeling.
- Built **NLP and text analytics models** using Transformers, Sentence Transformers, and BERT for market sentiment analysis, semantic search, NER, topic modeling, and summarization to extract consumer insights from unstructured data.
- Designed and maintained scalable **ETL/ELT pipelines** using Apache Airflow, Apache Spark, PySpark, and dbt; integrated Snowflake data warehousing and Delta Lake, reducing data processing time by **25%** and ensuring reliable data availability for downstream ML models.
- Deployed ML models as **RESTful APIs via FastAPI and Docker** with real-time Kafka and PySpark streaming inference pipelines; implemented model monitoring, drift detection, and automated retraining workflows using MLflow and CI/CD pipelines.
- Applied SHAP and LIME for model explainability, built **Prometheus and Grafana** observability dashboards cutting deployment time by **20%**, and delivered Power BI and Tableau insights for 10+ global markets in Agile sprint cycles.

Technologies Used: XGBoost, LightGBM, PyTorch, TensorFlow, BERT, Transformers, FastAPI, Docker, Kafka, PySpark, Apache Airflow, Apache Spark, dbt, Snowflake, Delta Lake, MLflow, Prometheus, Grafana, SHAP, LIME, Power BI, Tableau

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Machine Learning Engineer | Maxton Technology Pvt. Ltd. | Bangalore, India | 08/2022 – 08/2023

Maxton Technology is a technology solutions company specializing in data engineering, machine learning, and enterprise AI development for clients across multiple industry verticals.

- Developed supervised and unsupervised **ML models** (Scikit-learn, XGBoost, CNNs, LSTMs) for demand forecasting, anomaly detection, and clustering, improving model accuracy by **15%** through iterative feature engineering, statistical modeling, and hyperparameter optimization.
- Applied **Computer Vision** techniques using OpenCV, CNNs, Vision Transformers (ViT), CLIP, SAM, and OCR for image classification, object detection, document intelligence, and automated data extraction workflows.
- Built **NLP models** for text classification, NER, topic modeling, and sentiment analysis; deployed production models via Flask, Docker, and RESTful APIs on GCP Vertex AI with cross-validation using AUC, RMSE, and F1.

- Designed scalable **ETL/ELT pipelines** using Python, PySpark, and Apache Spark on GCP (Vertex AI, BigQuery, Cloud Storage), reducing pipeline latency by **25%**, and ensured data integrity through automated validation and version-controlled codebases in Agile environments.

Technologies Used: Scikit-learn, XGBoost, CNNs, LSTMs, OpenCV, ViT, CLIP, SAM, Flask, Docker, GCP Vertex AI, BigQuery, Apache Spark, PySpark, Python, spaCy, HuggingFace

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AI/ML Intern | Venture Offshore Infomatrix Pvt. Ltd. | Vishakhapatnam, India | 02/2022 – 07/2022

Venture Offshore Infomatrix is a technology services and software development company focused on insurance SaaS platforms and AI-driven automation solutions.

- Developed **Python-based ETL pipelines** to ingest and transform structured insurance policy and claims data, reducing manual data processing time by **20%**.
- Built and fine-tuned **NLP models (BERT, spaCy)** for insurance document classification, named entity recognition (NER), and claims text summarization, improving processing accuracy by **25%**.
- Assisted in building **LLM-powered chatbot workflows** using HuggingFace Transformers for automated customer query resolution, reducing manual support effort by **30%**.
- Deployed **ML models as microservices** using Docker and FastAPI, enabling scalable AI-native insurance SaaS platform integration.
- Designed and integrated **RESTful APIs** to enable seamless data exchange between AI modules and insurance platform components; created interactive dashboards using Streamlit and Power BI to visualize claims trends and policy KPIs.

Technologies Used: Python, BERT, spaCy, HuggingFace Transformers, FastAPI, Docker, PostgreSQL, Streamlit, Power BI, Git, ETL Pipelines, REST APIs

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FEATURED PROJECTS

Enterprise AI Knowledge Retrieval System (RAG + Agentic AI)

- Architected production RAG and GraphRAG system for 100,000+ enterprise documents with hybrid semantic and sparse retrieval, cross-encoder reranking, chain-of-thought reasoning, and Knowledge Graphs (Neo4j), achieving sub-second latency, 35% accuracy improvement, and 99.9% uptime.
- Built LangGraph multi-agent Agentic AI with MCP tool integration, supervisor-worker patterns, autonomous task decomposition, human-in-the-loop controls, and Guardrails AI, deployed via FastAPI, Kubernetes (EKS), ArgoCD, GitOps, OpenTelemetry, and Terraform IaC on AWS and Azure.
- **Tech Stack:** LangChain, LangGraph, FAISS, Pinecone, Weaviate, Azure OpenAI, GraphRAG, Neo4j, FastAPI, Docker, Kubernetes, Terraform, OpenTelemetry

LLM Fine-Tuning and Multi-Agent Orchestration Pipeline

- Built end-to-end LLM fine-tuning pipeline with LoRA, QLoRA, PEFT, RLHF, DPO, quantization, and distillation, reducing compute cost by 60%; evaluated with RAGAS and TruLens, tracked via MLflow with drift detection and automated retraining triggers.
- Designed multi-agent orchestration framework using LangGraph, CrewAI, and AutoGen with tool calling, session-aware memory, fault tolerance, and human-in-the-loop, deployed via vLLM inference, Datadog observability, and Docker/Kubernetes services.
- **Tech Stack:** LoRA, QLoRA, PEFT, RLHF, DPO, PyTorch, HuggingFace, LangGraph, CrewAI, AutoGen, RAGAS, TruLens, MLflow, vLLM, Datadog

Demand Forecasting ML System & Full MLOps Pipeline

- Built scalable demand forecasting engine using XGBoost, Neural Networks (PyTorch), and ensemble methods with automated feature engineering, A/B testing, and drift-triggered retraining, achieving 20% prediction accuracy improvement deployed on AWS SageMaker.
- Implemented full MLOps pipeline with MLflow, Apache Airflow, dbt, Docker, Kubernetes, CI/CD (GitHub Actions), real-time Kafka streaming for feature generation, Snowflake feature store, and Power BI dashboards for SLA and pipeline monitoring.
- **Tech Stack:** XGBoost, PyTorch, Apache Airflow, MLflow, Snowflake, Kafka, AWS SageMaker, Docker, Kubernetes, Power BI

EDUCATION

Master of Science in Artificial Intelligence | Saint Louis University, St. Louis, MO | 08/2023 – 12/2025